

AI Ecosystem for Fake Job Scam Detection and Fraud-Aware Job Recommendation

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Project Proposal Report

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Declaration

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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Abstract

Online job platforms have significantly improved the efficiency of the hiring process by allowing companies to post job opportunities and connect with potential candidates digitally. However, the rapid growth of online employment platforms has also led to an increase in recruitment fraud, in which scammers impersonate legitimate organizations or create fake job posts to mislead job seekers. Most current research pays little attention to verifying the legitimacy of the employer behind a job opportunity, instead focusing on examining the textual content of job postings to identify fraudulent listings. Because of this, dishonest recruiters can still exploit these platforms by crafting job postings that appear professional while concealing themselves behind dubious or inconsistent employer identities. This study suggests an Employer Legitimacy Verification System to close this gap. Instead of depending just on job-advertisement analysis, the system assesses employer credibility using company-level data. Employer-related data, including firm name, website domain, recruiter email domain, contact information, LinkedIn presence, and employer profile qualities, are gathered by the proposed system. These characteristics are converted into structured credibility indicators that reflect employer identification transparency and consistency through feature engineering. Employers are then categorized as either authentic or possibly dubious using machine-learning classification techniques such as Random Forest, Logistic Regression, and XGBoost. Additionally, the main factors influencing the model's prediction are identified and presented using SHAP-based explainable AI techniques, thereby increasing user confidence and transparency. Python, Scikit-learn, XGBoost, Pandas, and React Native will be used in the system's implementation, and a backend API will link the machine learning model to a mobile application interface. A trustworthy employer verification system that helps job seekers identify dubious employers before applying for positions is the anticipated result. The suggested method can help reduce recruitment fraud and promote safer digital employment ecosystems by enhancing transparency and confidence in online recruitment.

Keywords: Recruitment Fraud Detection, Employer Legitimacy Verification, Machine Learning, Random Forest, Logistic Regression, XGBoost, Explainable AI, Online Recruitment Security

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List of Abbreviations

Table 1 - List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
ML	Machine Learning
NLP	Natural Language Processing
XAI	Explainable Artificial Intelligence
SHAP	Shapley Additive Explanations
RF	Random Forest
LR	Logistic Regression
XGBoost	Extreme Gradient Boosting
URL	Uniform Resource Locator
REST	Representational State Transfer
NoSQL	Not Only Structured Query Language
SDG	Sustainable Development Goal
UI	User Interface
API	Application Programming Interface

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1. Introduction

1.1 Background and Context

Online recruitment platforms have changed the hiring process. They allow organizations to post job openings and connect with potential candidates online. The widespread use of job portals and recruitment systems has made hiring more efficient and increased job opportunities for job seekers worldwide. However, the rapid growth of online recruitment has also led to more fraud. This includes online recruitment fraud, or fake job scams, where attackers pose as legitimate employers or create false job ads to mislead job seekers [1], [2]

In today's digital environment, recruitment scams are a significant cybersecurity issue. Online methods are sometimes used by unreliable recruiters to reach out. Recruitment fraud is one of the most significant cybersecurity problems in today's digital world. Online platforms are often used by unreliable recruiters to reach out to candidates or post fictitious job postings. They want personal information, registration fees, or training expenses in exchange for job opportunities. Vulnerable groups, such as young graduates, jobless individuals, and those seeking remote work, are targeted by these frauds. According to research, phony job postings can result in monetary losses and reveal victims' private information, undermining consumers' confidence in online job boards [3], [4].

To address this issue, researchers have explored various ML and data mining methods to detect fake job postings and recruitment scams. Several studies have proposed classification models that examine job advertisement features such as job descriptions, job titles, salary information, and other textual attributes to identify questionable job postings [5], [6], [7]. ML algorithms such as Random Forests, Logistic Regression, Support Vector Machines, and numerous other classification methods have been used in several studies to model potential advertising fraud by leveraging patterns identified in the data [8], [9].

Despite these advancements, many existing approaches focus primarily on examining job advertisement content rather than assessing the credibility of the employer behind the job posting. Fraudulent recruiters can often create convincing job ads that seem legitimate while masking suspicious company identities. As a result, systems that rely solely on job advertisement analysis might miss scams by fake or unverified employers [10], [11].

To address these limitations, this research introduces an **Employer Legitimacy Verification Model** that evaluates the credibility of employers behind job postings. Instead of analyzing only job-post text, the proposed approach examines employer-related attributes and indicators of identity consistency. These include the consistency between the company name and website domain, recruiter email domain alignment with the official company domain, availability of verifiable company contacts information, LinkedIn company presence, and overall employer profile completeness. Through feature engineering, these attributes are transformed into structured credibility indicators that signal employer transparency and identity consistency. The proposed approach applies feature engineering techniques to transform raw employer data into structured credibility indicators. These indicators are then analyzed using machine learning classification models, particularly Random Forest, to

categorize employers as legitimate or potentially suspicious. The proposed approach applies feature engineering techniques to transform raw employer data into structured credibility indicators. These indicators are then analyzed using machine learning classification models, particularly Random Forest, to categorize employers as legitimate or potentially suspicious.

Protecting job seekers from fraud is crucial for a fair digital economy. This research focuses on that vulnerability by checking the legitimacy of employers on recruitment platforms before anyone submits an application. By supporting SDG 8: **Decent Work and Economic Growth**, our work goes beyond just detection. We use Explainable AI (XAI) to make sure the "why" behind a suspicious flag is as clear as the "what." This project is a key part of our Artificial Intelligence and Data Analytics group.

1.2 Literature Review

Overview of Existing Research

The rapid growth of online recruitment platforms has created new opportunities for both employers and job seekers. However, it has also led to a significant rise in recruitment scams and fake job advertisements. In response to this issue, many researchers have explored using ML, natural language processing, and data mining techniques to detect fraudulent job postings and recruitment scams. Most existing research studies analyze job post data, including job descriptions, job titles, salary information, and other posting-related attributes, to identify suspicious patterns associated with recruitment scam [5], [6], [7], [1]

Recent research shows that ML algorithms such as Random Forest, Logistic Regression, Support Vector Machines, and ensemble learning methods can effectively classify job postings as legitimate (real) or fraudulent (fake) [5], [6] [7]. Other studies also focused on contextual and behavioral indicators related to recruitment fraud, highlighting the importance of combining multiple data features to improve detection performance [1], [3]. Despite these advances, most existing research focuses on analyzing job ad content rather than evaluating the credibility of the employer behind the posting.

Critical Literature Review Table

Table 2 - Literature Review

Study	Research Focus	Data / Features Used	Methods / Algorithms	Key Contribution	Limitation
Alghamdi & Alharby, 2019 [1]	Intelligent model for recruitment fraud detection	Job metadata and textual features	ML techniques	Demonstrates ML effectiveness for recruitment	Focus remains on job advertisement analysis

				fraud detection	
Vidros et al. (2017) [2]	Automatic detection of online recruitment fraud and analysis of fraudulent job postings	Public dataset of job advertisements including textual features	ML classification techniques for fraud detection	Introduced a publicly available dataset for fake job detection and analyzed the characteristics of fraudulent recruitment posts	Focuses mainly on job advertisement content rather than verifying the legitimacy of the employer or company behind the posting
IJETT Fake Job Detection Study [3]	Detection of online employment scams	Job advertisement text and metadata	ML classification	Identifies patterns in fraudulent job postings	Employer identity and company credibility are not considered
Ansar & Hussain (2025) [4]	Impact of online job scams on users' trust in digital platforms	Survey data and user perception regarding online job scams	Statistical and survey analysis	Highlights how recruitment scams reduce trust and reliance on online job platforms	Does not propose a technical detection or prevention model for identifying fraudulent employers or job postings
Allam et al., 2025 [5]	Detection of fake job postings using ML	Job advertisement features, recruitment data	Data balancing + ML classifiers	Improved fake job detection accuracy using balanced datasets	Focuses mainly on job advertisement data rather than employer credibility
Vijayalakshmi et al., 2024 [6]	Fake job post identification using Python-based ML	Job description text and posting attributes	Elevated learning strategy, ML classification	Demonstrates effective classification of fraudulent job postings	Does not evaluate employer legitimacy indicators
Medapati et al., 2025 [7]	ML approach for fake	Job advertisement features	ML classifiers	Identifies suspicious job postings using	Limited focus on employer-

	recruitment detection			classification models	level verification
IRJMETs Fake Job Detection Study [8]	Fake job identification using ML	Job description features	Random Forest classification	Shows the strong performance of Random Forest in detecting fake jobs	Heavy reliance on textual job features
JETIR Recruitment Scam Study [9]	Online employment scam detection	Job advertisement data	ML classification techniques	Demonstrates data mining methods for scam detection	Limited focus on employer credibility indicators
SSRN Fake Job Advertisement Study [10]	Detection of fake job advertisements	Job advertisement attributes	ML models	Demonstrates classification- based fraud detection	Limited explainability and employer- level indicators
SSRN Job Scam Detection Study [11]	Online job scam detection using ML	Recruitment patterns	ML classifiers	Highlights the effectiveness of ML in recruitment fraud detection	Does not verify employer authenticity
IJCRT Fake Job Detection Study [12]	Fraudulent job detection using ML	Job posting metadata	ML models	Identifies fraudulent patterns in job advertisements	Focus limited to advertisement- level analysis

1.3 Research Gap

Even though a lot of research has been done on detecting recruitment fraud on online job platforms, most studies now focus more on **identifying fraudulent job postings than on verifying the employer's legitimacy**. Many researchers have applied ML algorithms such as Random Forest, Logistic Regression, and other classification techniques to analyze job posts data, including job descriptions, titles, salary information, and posting attributes [5], [6], [7].

However, relatively limited attention has been given to evaluating the **credibility of the employer** associated with the job advertisement. In many cases, fraudulent recruiters create professional-looking job posts that appear legitimate while concealing suspicious or inconsistent employer identities. As a result, systems that rely primarily on job-advertisement analysis may fail to detect scams originating from **fake or impersonated employers**.

Additionally, employer-related credibility indicators, such as consistency between the company name and website domain, alignment of recruiter email domains with the official company domain, availability of verifiable company contact information, and completeness of employer profile information, are rarely systematically analyzed in existing detection systems. This creates a gap in current research, as employer legitimacy verification can provide an additional layer of protection against recruitment fraud.

Furthermore, many existing systems provide only a **binary classification (e.g., fake or legitimate) without clearly explaining the factors that influence** the prediction. This lack of transparency reduces user trust and makes it difficult for job seekers to understand why a particular employer may be considered suspicious.

To address these limitations, this research proposes an **Employer Legitimacy Verification Model** that evaluates employer credibility using multiple employer-level indicators derived from feature engineering. The proposed system applies machine learning algorithms to analyze these indicators and classify employers as legitimate or potentially suspicious. In addition, explainable AI techniques such as SHAP are used to identify and interpret the key indicators that influence the model's predictions, thereby improving transparency and user trust in the system's decision-making process.

2. Objective

2.1 Main Objective

To design and develop an explainable ML-based Employer Legitimacy Verification System that analyzes company-level employer credibility indicators and identity consistency features to classify employers as legitimate or potentially suspicious, while providing risk indicator highlighting and verification confidence levels to improve transparency and user trust in online recruitment platforms.

2.2 Specific Objectives

1. To identify and analyze employer credibility indicators relevant for verifying employer legitimacy, including consistency between company name and website domain, recruiter email domain alignment with the official company domain, availability of verifiable company contacts information, LinkedIn company presence, and employer profile completeness.
2. To collect and construct an employer dataset by gathering publicly available company information from job postings and online company sources, and applying feature engineering techniques to generate structured machine learning features.
3. To develop and train machine learning classification models, particularly Random Forest, to analyze employer credibility indicators and classify employers as legitimate or potentially suspicious.
4. To apply explainable AI techniques (SHAP) to identify and highlight the key credibility indicators that influence the model's predictions, improving transparency and interpretability.
5. To integrate the employer legitimacy verification model into the backend of the system, enabling users to verify employer credibility and view explanations for suspicious employer classifications.
6. To evaluate the effectiveness of the proposed model using machine learning performance metrics such as accuracy, precision, recall, and F1-score, and to assess the system's usefulness in supporting safer job application decisions.

3. Methodology

Research Approach

This research follows a **systems development and evaluation methodology** commonly used in information technology research to design, implement, and assess a technological solution to a real-world problem. The purpose of this research is to design and develop an Employer Legitimacy Verification Model that can identify suspicious employers in online job platforms using company-level trust indicators.

The research process includes several stages: data collection, dataset creation, feature engineering, model development, and performance evaluation. In this approach, employer-related information is collected and transformed into structured ML features. A classification model (Random Forest) is then trained on the collected dataset, and its performance is evaluated using standard metrics. This approach allows the proposed model to be tested and validated through controlled experiments.

Technical Methods and Algorithms

Dataset Construction

An employer-level dataset is created by gathering employer data from online job posts and publicly accessible firm sources to develop the Employer Legitimacy Verification Model. Employer features, including company name, website URL, recruiter email address, contact details, LinkedIn company presence, and employer profile details, are all included in the dataset.

Feature engineering is used to create additional credibility indicators from these basic attributes. These indicators include the availability of verifiable contact information, a LinkedIn company presence, employer profile completeness, domain consistency between the company name and its website domain, and alignment between the recruiter email domain and the company website domain.

Based on available verification evidence and dataset annotations, each employer instance in the dataset is classified as either legitimate or suspicious. Structured employer credibility indicators that are used as input features for ML classification are included in the final dataset.

Feature Engineering

Feature engineering is applied to transform the collected employer data into structured credibility indicators suitable for machine learning analysis.

The feature engineering process includes the following steps:

- **Domain Extraction:** Website domains and recruiter email domains are extracted from URLs and email addresses.

- **Identity Consistency Analysis:** Similarity between the company name and the website domain is computed using fuzzy string-matching techniques.
- **Domain Alignment Verification:** The recruiter's email domain is compared with the official company website domain to determine domain consistency.
- **Profile Completeness Calculation:** A profile completeness score is calculated based on the proportion of required employer attributes that are available.
- **Binary Indicator Generation:** The presence of employer information, such as contact details, physical address, and LinkedIn company page, is converted into binary features.

These engineered indicators signal employer transparency, identity consistency, and digital presence.

Model Architecture

To classify employer legitimacy, this research applies multiple machine learning algorithms to analyze indicators of employer credibility. The selected algorithms include **Random Forest, Logistic Regression, and XGBoost**.

These models are trained using engineered employer credibility features, such as domain consistency between the company name and the website domain, recruiter email domain alignment, employer profile completeness, LinkedIn company presence, and the availability of verifiable contact information.

Random Forest:

Random Forest is used as the **primary classification model** in this research. It is an ensemble learning algorithm that constructs multiple decision trees during training and aggregates their predictions to produce the final classification result. Random Forest is selected because it performs well with structured tabular datasets and can capture complex relationships between multiple credibility indicators while reducing the risk of overfitting.

Logistic Regression:

Logistic Regression is used as a **baseline classification model**. This algorithm models the probability of a binary outcome using a logistic function. It is widely used in classification problems due to its simplicity, interpretability, and ability to provide probabilistic outputs. Logistic Regression allows comparison with more complex models to evaluate whether the proposed features improve prediction performance.

XGBoost:

XGBoost (Extreme Gradient Boosting) is also applied as an additional classification model. XGBoost is a powerful gradient boosting algorithm that builds sequential decision trees to improve prediction accuracy. It is widely used in structured data problems due to its efficiency, scalability, and strong predictive performance.

Model Comparison

The performance of these models is evaluated and compared using standard classification metrics such as **accuracy, precision, recall, and F1-score**. Comparing multiple algorithms allows the researcher to determine which model best identifies suspicious employers based on credibility indicators.

Explainability Layer

To improve transparency and interpretability, the system uses **SHAP** to analyze each credibility indicator's contribution to the model's prediction.

SHAP identifies the most influential employer credibility features affecting each prediction, such as:

- email domain mismatch with company domain
- low employer profile completeness
- missing LinkedIn company presence
- inconsistent company name and website domain

These explanations are converted into human-readable messages and presented to users within the system interface. This allows job seekers to understand the key risk indicators associated with suspicious employers.

System Integration

The trained machine learning model is integrated into the backend system and connected to a **cross-platform mobile application developed using React Native**.

Through the mobile application, users can enter employer information such as company name, website URL, and recruiter email address. The application sends requests to the backend system via RESTful APIs, where feature extraction and prediction are performed.

The backend returns the **employer's legitimacy classification, confidence score, and explanation results, allowing users to verify the employer's** credibility before applying for job opportunities.

Tools and Platforms

Table 3 - Tools and Platforms

Tool / Technology	Purpose	Justification
Python	Machine learning development and data processing	Python is widely used in data science and machine learning due to its extensive ecosystem of libraries for data analysis, feature engineering, and predictive modeling.
Scikit-learn	Implementation of Random Forest and	Scikit-learn provides reliable and efficient implementations of machine learning

	Logistic Regression models	algorithms and is well suited for classification tasks on structured datasets.
XGBoost Library	Implementation of the XGBoost classification model	XGBoost is a powerful gradient boosting framework known for high performance and efficiency in structured data classification problems.
SHAP (Shapley Additive Explanations)	Model explainability and feature importance analysis	SHAP provides interpretable explanations by measuring the contribution of each feature to the model's prediction, improving transparency and user trust.
Pandas & NumPy	Data preprocessing, feature engineering, and dataset manipulation	These libraries enable efficient handling of structured datasets, allowing preprocessing operations such as cleaning data, handling missing values, and generating credibility indicators.
RapidFuzz / Fuzzy Matching	String similarity analysis for identity consistency verification	RapidFuzz is used to compute similarity between company names and domain names, enabling detection of identity inconsistencies between employer information sources.
React Native	Development of cross-platform mobile application	React Native enables the development of a single mobile application that can run on both Android and iOS devices, improving accessibility for users.
Node.js / Express.js or Flask API	Backend service for integrating the ML model with the mobile application	These frameworks allow the creation of lightweight RESTful APIs that enable communication between the mobile application and the machine learning model.
MongoDB	Database for storing application and verification request data	MongoDB is a flexible NoSQL database capable of storing structured and semi-structured data such as employer verification requests and prediction results.
Jupyter Notebook / Google Colab	Experimental environment for model development and evaluation	These platforms provide interactive environments for data exploration, model experimentation, and visualization during the research process.
Matplotlib / Seaborn	Visualization of model performance and feature importance	These libraries support the visualization of model evaluation results, feature importance, and performance metrics to support analysis and reporting.

Data Collection and Ethical Consideration

Employer-related data gathered from publicly accessible sources, including business websites, online job boards, and domain registration databases, will make up the dataset used in this study. Indicators of the legitimacy of an employer's web presence will be included in the collected data.

During the data collection, ethical considerations will be carefully followed. Only publicly accessible information will be used, and no personal or sensitive data will be collected without permission. The information gathered will be used solely for scholarly research, and confidentiality and privacy will be maintained at all times.

Validation Strategy and Performance Metrics

A training-testing dataset split, in which a portion of the dataset is used to train the model and the remainder to test its performance, will be used to assess the proposed model. This technique guarantees that the model can effectively generalize to new data.

The model's efficacy will be assessed using a number of categorization performance criteria, such as:

- Accuracy: The overall correctness of the model's predictions is measured.
- Precision: Quantifies the proportion of anticipated suspicious employers who are, in fact, suspicious.
- Recall: Assesses how successfully the model detects all suspicious employers
- F1-Score: Offers a fair assessment of recall and precision.

The effectiveness of the employer legitimacy verification approach in identifying suspicious employers and enhancing recruitment fraud detection will be assessed using these measures. In addition to classification performance metrics, the usability of the explanation outputs, risk indicators, and verification confidence levels will be evaluated through a user study using a Likert-scale questionnaire.

User Evaluation

A Likert-scale questionnaire will be used in user research to evaluate the system's usability and reliability. At least fifty people will be given the survey, where consumers will assess:

- Usefulness of the system
- Trust in the legitimacy verification results
- Clarity of the explanations provided by the system
- Usefulness of confidence indicator
- Usefulness of risk highlighting

4. High-Level System Architecture

Overview of the Proposed Solution and Its Components

An Employer Legitimacy Verification Model is the suggested remedy, intended to determine whether an employer on a web job site is trustworthy or questionable. Rather than depending solely on the text of job posts, the system analyzes trustworthiness factors at the organization level.

This proposed architecture contains five main components:

1. **Mobile Application Interface**
Provides the front-end interface for users to enter employer details and view legitimacy results.
2. **API / Backend Service**
Handles communication between the mobile application and the ML model.
3. **Data Collection Module**
Collects employer-related information such as company website, email domain, domain age, SSL availability, and social media presence.
4. **Feature engineering and Preprocessing Model**
Converts raw employer information into structured features suitable for ML.
5. **Employer Legitimacy Verification Model**
Classifies employers as legitimate or maybe suspect using a Random Forest classifier.
6. **Explainability Module**
Uses SHAP to identify key employer indicators that influence the prediction and to highlight potential risk factors.
7. **Database (MongoDB)**
Stores employer verification requests, processed data, prediction results, and system logs.
8. **Verification Confidence Module**
Stores employer verification requests, processed data, prediction results, and system logs.

System Diagram of the Components

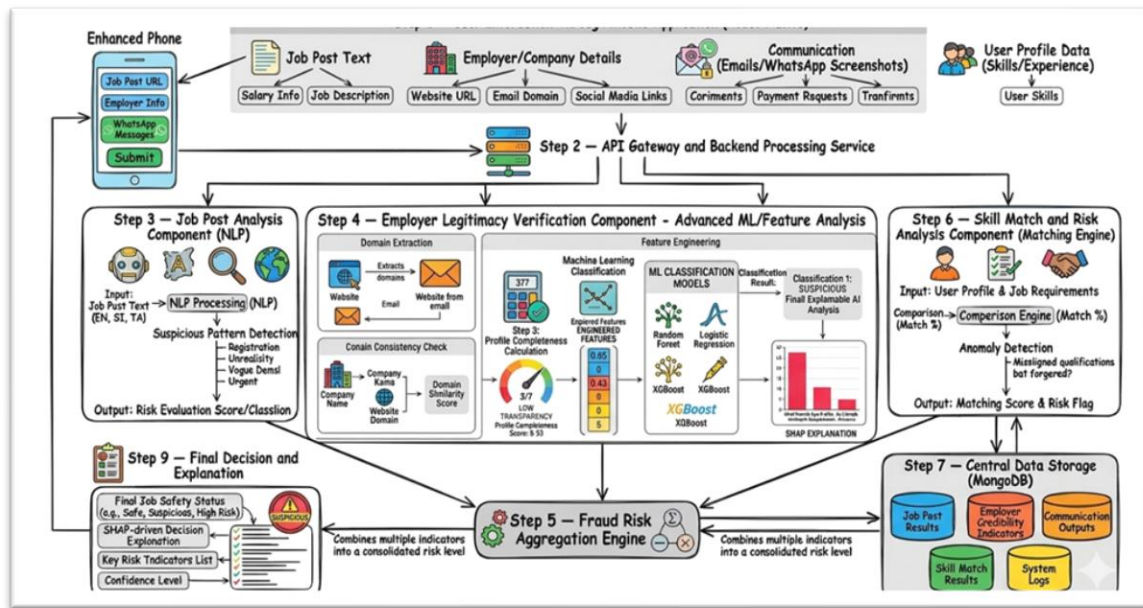


Figure 1 - System Diagram of the components

Architecture Diagram for Individual Component

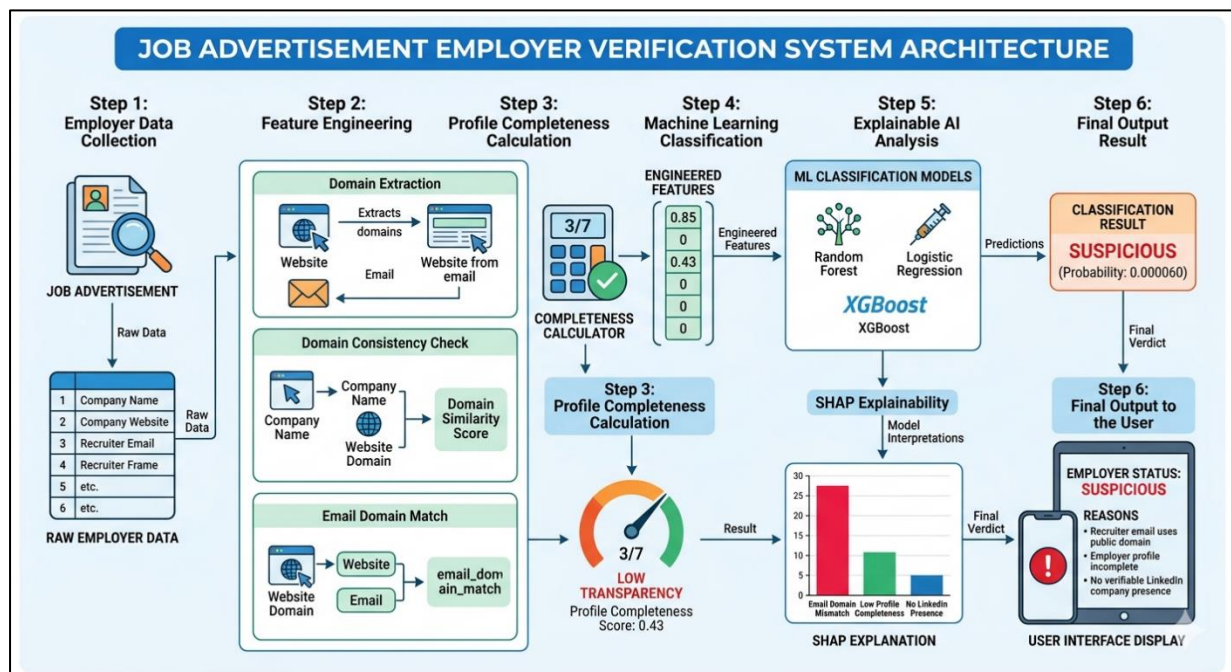


Figure 2 - Architecture Diagram for Individual Component

Component Interactions

Each component works together in sequence:

- First, the user interacts with the Mobile Application Interface by entering employer-related details for verification.
- The request is then sent to the API / Backend Service, which manages communication between the mobile application, database, and machine learning model.
- The Data Collection Module gathers employer-related information from publicly available sources, such as company websites, domain services, and online company profiles.
- This information is passed to the Feature Engineering and Preprocessing Module, where raw employer data is cleaned, transformed, and converted into structured machine learning features.
- The processed features are sent to the Employer Legitimacy Verification Model, where machine learning algorithms (Random Forest, Logistic Regression, and XGBoost) analyze the credibility indicators to classify the employer as legitimate or suspicious.
- The model output is passed to the Verification Confidence Module, which generates a confidence level indicating how reliable the prediction is.
- At the same time, the prediction and its associated feature values are forwarded to the SHAP Explainability Module, which identifies the employer credibility indicators that had the greatest influence on the model's decision. These indicators are presented as risk indicators when suspicious patterns are detected.
- The final prediction result, confidence level, and explanation are stored in the MongoDB Database for system logging, result management, and future reference.
- Finally, the API / Backend Service returns the legitimacy decision, confidence indicator, and explanation to the Mobile Application Interface, where the user can view the employer verification result in a clear and understandable format.

Data Flow Explanation

The following is a description of the data flow in the suggested system:

- Employer information is gathered from publicly accessible websites.
- After cleaning, the raw data is converted into organized machine-learning features.
- The Random Forest model receives the processed feature set.
- The employer's legitimacy class is predicted by the model.
- SHAP evaluates the forecast and pinpoints the key contributing factors.
- The user is shown the ultimate legitimacy ruling and justification.
- This flow enhances the system's usefulness and transparency by supporting both prediction and interpretability.

Scalability and Security Considerations

Scalability:

The proposed architecture is scalable because each module performs a separate task and can be extended independently. For example:

- Additional employer indicators could be added to the dataset.
- Larger datasets can be used to retrain the model.
- To accommodate more user requests, the backend can be scaled.
- In the future, further verification modules might be added.

This modular design allows the system to support future enhancements and increasing user demand.

Security:

Because the system analyzes employer-related data and predicts fraud, security is crucial. The following actions are taken into consideration:

- Only publicly available employers will be used to create the dataset.
- No sensitive personal user data will be collected unnecessarily.
- Communication between the mobile app and backend can be secured through HTTPS.
- Access to stored data and model services will be restricted during development and deployment
- Data handling will follow ethical and responsible research practices.

These factors help ensure the suggested method is safe, dependable, and appropriate for use in online hiring settings.

5. User Requirements

Stakeholder Identification

The suggested system interacts with or benefits several stakeholders. The Employer Legitimacy Verification System is used and developed by these stakeholders in various ways.

The suggested system interacts with or benefits several stakeholders. The Employer Legitimacy Verification System is used and developed by these stakeholders in various ways.

Table 4 - Stakeholder Identifications

Stakeholder	Description
Job Seekers	Primary users who verify employer legitimacy before applying for jobs.
Recruitment Platforms	Platforms that may integrate the verification system to reduce recruitment fraud.
System Developers	Responsible for developing and maintaining the mobile application and backend model.
Researchers / Administrators	Monitor system performance, update datasets, and retrain the ML model when needed.

As the system seeks to enhance their safety when interacting with online job opportunities, job seekers are the primary target users among these stakeholders.

Functional Requirements

The following essential features must be offered by the system:

1. Through the mobile application, users will be able to enter employer or company information.
2. Employer-related information, including company name, company website URL, recruiter email address, company contact information, LinkedIn company presence, and employer profile details, will be collected and processed by the system.
3. To transform unprocessed business data into ML features, the system will use feature engineering.
4. The Employer Legitimacy Verification Model will be used by the system to examine employer data.
5. Employers will be categorized by the system as either Potentially Suspicious Employer or Legitimate Employer.
6. Using SHAP-based explainability approaches, the system will explain the classification.
7. Using the mobile application interface, the system will show the user the legitimacy result and an explanation.

8. When new employer data becomes available, the system must enable the backend model to be updated or retrained.

Non-Functional Requirements

The non-functional requirements focus on features that ensure the system is reliable, comprehensible, and practical for verifying employer authenticity.

1. **Explainability:**

The SHAP explainability framework will be used by the system to produce comprehensible explanations for every forecast. To help users understand why an employer appears legitimate or potentially suspicious, the system will display the top contributing indicators of employer credibility (such as domain age, email domain authenticity, and website presence) for each employer verification request.

2. **Reliability:**

Based on employer trustworthiness factors and a trained Random Forest model, the system will consistently produce legitimacy-verification findings. When processing comparable employer data inputs, the model must maintain consistent predictive performance with low variance under typical system operation, ensuring users receive dependable verification results.

3. **Performance:**

Under normal operating conditions, the system must provide an employer legitimacy categorization result within 2 seconds of receiving the employer information request. Users will receive verification results through the mobile application in a timely manner thanks to effective backend processing and lightweight API connectivity.

4. **Usability:**

In up to 3 interaction steps, users can enter employer information and receive verification results via the mobile application interface. To make it easy for users to understand the verification outcome, the system must provide validity results and SHAP-based explanations in a clear, user-friendly format.

5. **Data Integrity:**

Through preprocessing and feature engineering, the system will ensure accurate and consistent processing of employer-related data. To ensure that employer indicators such as domain age, email domain type, and website status are properly formatted and processed, input data will be reviewed prior to model prediction.

6. **Security:**

Using HTTPS protocols, the system will ensure secure communication between the mobile application and backend services. The system will collect and process

only publicly accessible employer data, ensuring that no private user information is misused or stored.

6. Commercialization Plan

Target Market and Customer Persona

The proposed system aims to provide job searchers and recruiting platforms with tools to verify the validity of companies before engaging in recruitment activities.

The primary target market includes:

- People who are actively looking for work via internet job boards.
- New graduates joining the workforce.
- Those looking for work remotely who mostly rely on job listings online.

The secondary market includes:

- Online employment platforms that intend to incorporate employer verification systems.
- Providers of job portals looking to increase platform security.
- Career guidance organizations that assist job seekers.

A new graduate or job seeker who often applies for jobs online and wants to confirm an employer's legitimacy before providing personal information or paying any registration fees is a common consumer persona.

Value Proposition

Before exploring job opportunities, job seekers can identify suspicious employers using the proposed system's mobile-based employer legitimacy verification service.

The system's key features include:

- Reducing the possibility of hiring fraud.
- Assisting users in immediately confirming the validity of employers via a mobile application.
- Delivering findings that are comprehensible and demonstrate the reasons behind judgments about legitimacy.
- Increasing safety and trust in online hiring environments.

The suggested approach offers automatic employer legitimacy verification, supported by ML and explainable AI techniques, in contrast to conventional employment platforms that rely on manual reporting or moderation.

Revenue Model

The suggested system can use several revenue models.

A premium model is one potential strategy in which ordinary users can access basic employer verification services for free, while premium subscribers can access more advanced features.

Possible sources of income consist of:

- Premium subscription packages for advanced verification capabilities.
- Access to APIs for hiring platforms looking to integrate employer verification services.
- Collaborations with career platforms and employment portals.
- Services at the corporate level for hiring firms

With this model, the system may reach a big user base and make steady profits.

Cost Estimation

The following are the main expenses related to the system's development and maintenance:

- Development and maintenance of mobile applications.
- Cloud hosting and backend server infrastructure.
- Training and maintenance of ML models.
- API services and data storage.
- Monitoring of systems and security.
- Open-source tools and cloud-based development environments can be used to create the system during the research phase, hence reducing the initial development costs.

Pricing Strategy

To accommodate various user categories, a tiered pricing strategy can be implemented.

Among the potential pricing choices are:

- Free Tier: fundamental validation of employer legitimacy.
- Premium Tier: comprehensive explanation outputs and sophisticated verification reports.
- Enterprise Tier: Integration of APIs for hiring platforms and companies.

The system can build a sizable user base and generate revenue from business or professional users under this pricing structure.

Competitive Advantage

Compared with current alternatives, the proposed system has several advantages.

- First, while the suggested method focuses on confirming the employer's legitimacy, most existing research and detection systems focus on identifying fake job postings.
- Second, the approach uses company-level credibility indicators that provide deeper insights into employer legitimacy, such as domain age, website existence, and email authenticity.

- Third, the use of explainable AI (SHAP) boosts user confidence in the system by enabling consumers to understand the rationale for the validity decision.
- Lastly, job seekers who rely on mobile devices for job searches will find the system useful and accessible if it is delivered through a mobile application.

Intellectual Property Considerations

The suggested approach involves developing a novel mobile application platform and an ML-based employer-verification mechanism.

Possible factors to take into consideration with regard to intellectual property include:

- Using software copyrights to safeguard the system architecture and model implementation.
- Safeguarding the source code of mobile applications
- Potentially submitting patent applications for innovative explainability integration or fraud detection techniques
- Additionally, by using publicly available firm information and refraining from collecting sensitive personal data, the system will adhere to ethical data-use principles.

7. Budget and Justification

Table 5 - Budget and Justification

Item	Cost (LKR)	Justification
Cloud Server (AWS / Google Cloud – 12 months)	20,000	Required to host the backend API and machine learning model used for employer legitimacy verification and mobile app communication.
MongoDB Atlas Cloud Database (12 months)	12,000	Used to store employer verification requests, prediction results, and application data generated by the mobile application.
Domain & API Services	8,000	Used to access domain-related information such as domain age, domain status, and SSL certificate verification.
Google Play Developer Account	9,000	Required to publish and distribute the mobile application on the Google Play Store for user access.
Survey Data Collection & Printing	7,000	Required for conducting the user evaluation study and printing survey questionnaires and research documentation.
Mobile App Testing Devices	10,000	Used for testing the React Native mobile application on multiple Android devices to ensure compatibility and performance.
Internet & Cloud Usage	6,000	Covers internet access, dataset downloads, and cloud experimentation during model training and system testing.
Miscellaneous Research Expenses	5,000	Includes documentation preparation, report printing,

		and other minor research-related expenses.
Total Estimated Cost:	77,000 LKR	

8. Work Breakdown Structure (WBS)

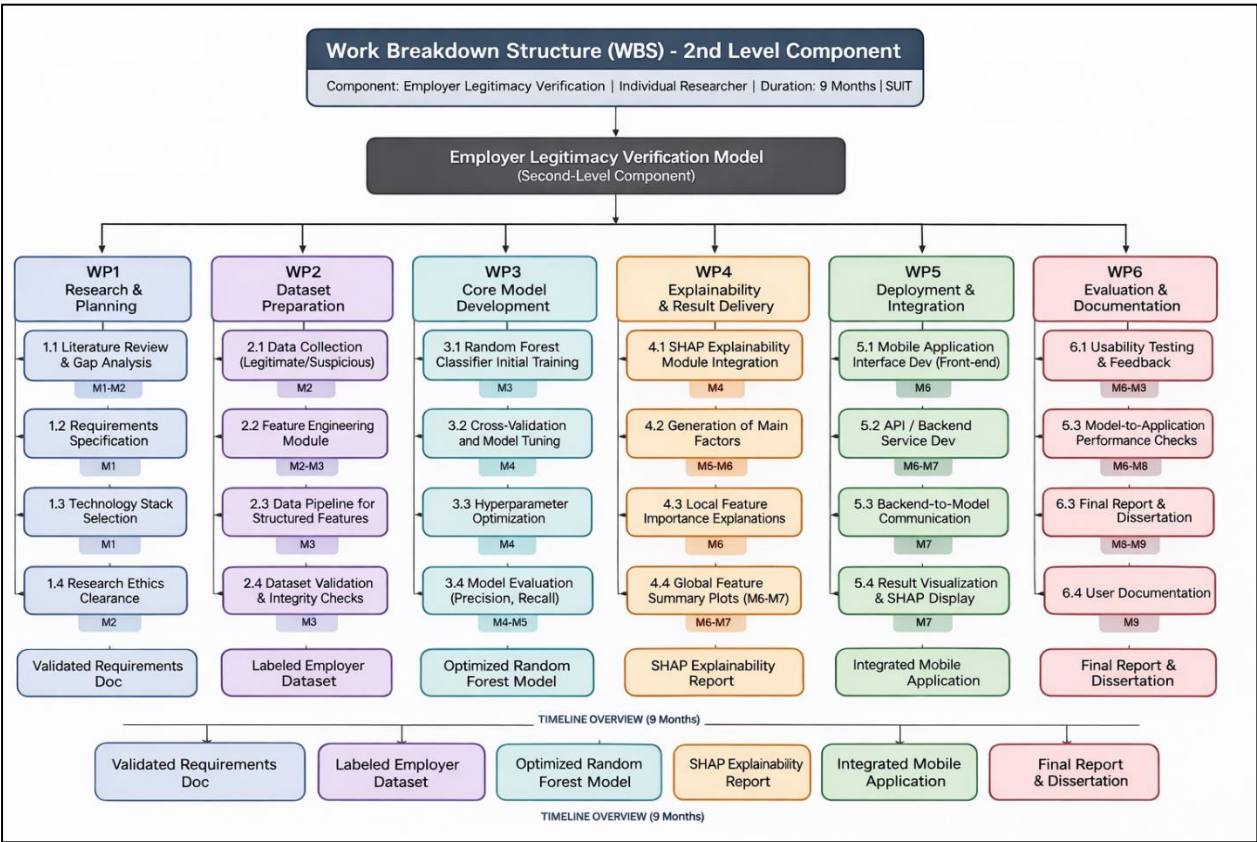


Figure 3 - Work Breakdown Structure

9. Gantt Chart

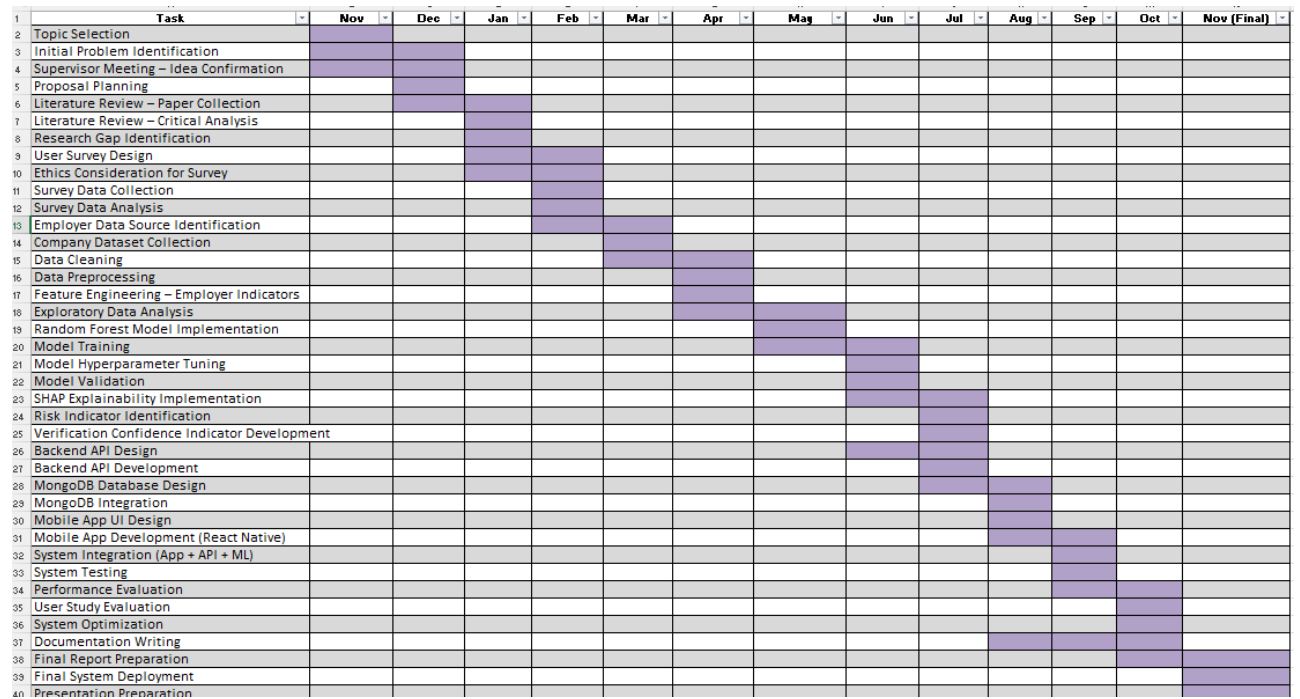


Figure 4 - Gantt Chart

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11. Appendices

Appendix A - Survey Instrument

The questionnaire used to gather user input on recruiting online experiences and employer verification requirements is included in this appendix. The purpose of the survey was to gather requirements for the planned employer-legitimacy verification system and to understand users' concerns about job scams.

The questions included:

- What is your group age?
- What is your current status?
- Have you ever searched for jobs online?
- Which platforms do you used to search for jobs?
- Have your ever seen fake or suspicious job advertisements online?
- What characteristics make a job advertisement look suspicious?
- Do you usually check the company details applying for a job?
- Which company-related factors increase your trust in a job opportunity?
- Have you ever received suspicious messages from someone claiming to be a recruiter?
- Through which platforms did you receive suspicious recruiter messages?
- What type of suspicious behavior did you notice in those messages?
- Have you ever been asked to pay money for a job opportunity?
- How confident are you in identifying fake job offers or recruitment scams?
- Do you think recruitment scams are increasing on online platforms?
- Would you use a system that automatically detects scam job messages or suspicious recruiters?
- Which features would help you detect job scams more effectively?
- Do you think AI technology can help reduce recruitment scams and protect job seekers?
- In your opinion, what is the biggest risk when applying for jobs online?

Appendix B - Preliminary Mobile Application UI Mockups

9:41

Verify Employer

Enter employer details below. Our AI analysis will check for legitimacy signals and risk indicators.

Company Name

e.g. Acme Tech Solutions

Company Website / URL

https://www.company.com

Job Posting URL

Link to the specific job ad

Contact Email/Phone

hr@company.com or +1...

Registration Number

e.g. CRN-12345678

OPTIONAL

Social Proof (LinkedIn)

linkedin.com/company/...

OPTIONAL

Additional Notes

Any red flags or context you've noticed...

OPTIONAL

Attach official document (PDF/JPG)

Run Verification

Need help? Tap the help icon at the top for tips on which data points improve verification accuracy.

SECURE AI VERIFICATION

Data is encrypted and used only for analysis. We do not share your verification history with third parties.

Home

Verify

History

Profile

Settings

9:41

Fake Job HireGuard

Employer Status: Legitimate Employer

Verification Confidence: High

TechSolutions Global

www.techsolutions.gl

Multiple independent records and verified company registration match. This employer has a consistent history of legitimate hiring processes.

VERIFIED SIGNALS

Official registration found

Working company website

Consistent contact domain

Verified job posting source

Learn more about our verification signals

Safety Verification

Our AI verification reduces the risk of job fraud by 98.4% for this specific employer profile.

Request Human Review

Watchlist

Report

VERIFICATION ID: HG-8821-X92

CHECKED ON OCT 24, 2023 • 14:32 PM

Home

Verify

History

Profile

Settings

9:41

Risk Indicators

How we verify

Our AI engine evaluates dozens of signals to determine the legitimacy of an employer. The indicators below explain the key signals used in our analysis.

Official Registration

Metric

We verify if the business is legally registered with national authorities like the SEC, Companies House, or local commerce bureaus.

EXAMPLE

"Cross-referencing the submitted registration number with the official UK Companies House database for an active status."

Website Match

Metric

Checks if the company's official web domain is consistent with established history and associated with the claimed business name.

EXAMPLE

"A recruitment link for 'Global Tech' should originate from globaltech.com, not a generic or misspelled 'global-tech-hr.org'."

Contact Domain

Metric

Analyzes the email address domain used for communication to ensure it belongs to the verified corporate entity.

EXAMPLE

"Authentic recruiters use corporate email like name@google.com rather than free services like @gmail.com or @outlook.com."

Job Posting Source

Metric

Evaluates the reputation and verification status of the platform where the job advertisement was originally discovered.

EXAMPLE

"Job posts on LinkedIn 'Verified' pages carry more weight than anonymous listings on local community classified boards."

Social Proof

Metric

Examines professional networks and social platforms to find consistent records of company activity, employees, and history.

EXAMPLE

"A legitimate firm typically has an active LinkedIn presence with employees who have verifiable work histories at that same firm."

Conflicting Records

Metric

Identifies discrepancies between submitted information and public records, such as mismatched addresses or phone numbers.

EXAMPLE

"The employer claims to be based in London, but the registration phone number is traced back to a known fraudulent VoIP origin abroad."

Still unsure about a result?

Our experts can perform a manual check for complex cases.

Ask Expert

"If unsure, request a human review for deeper investigation."

Home

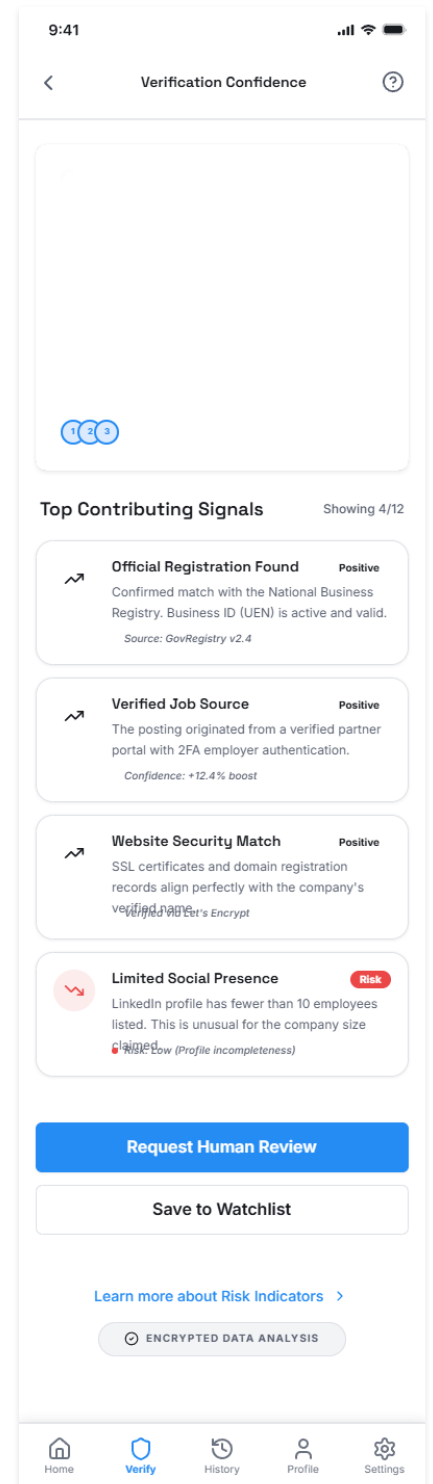
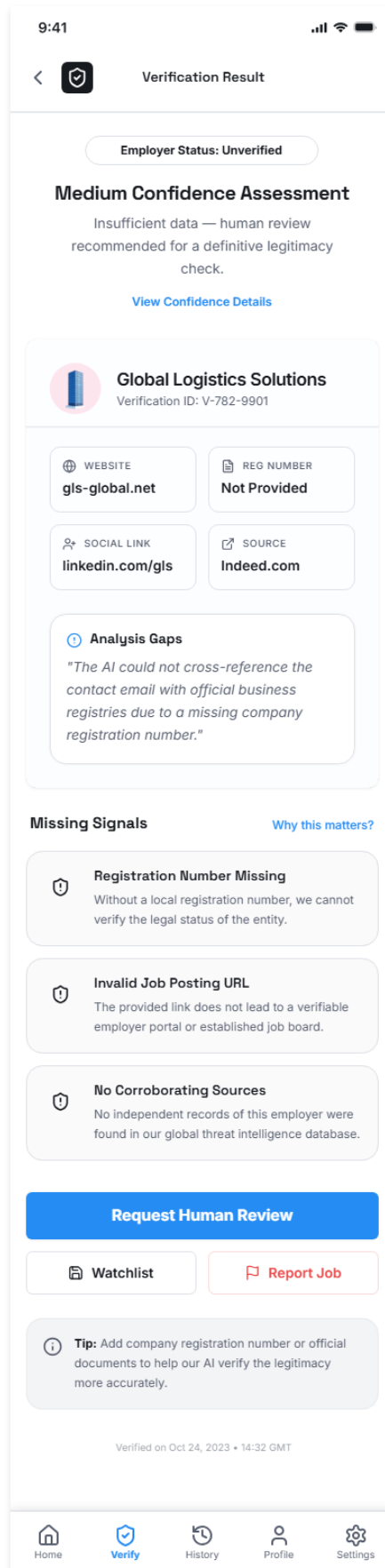
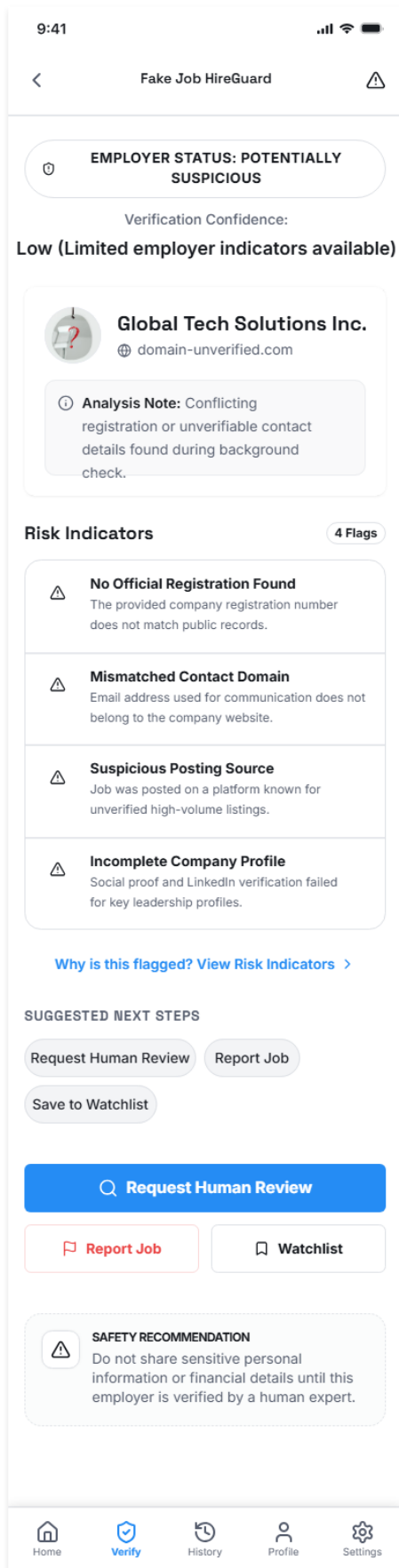
Verify

History

Profile

Settings

36



Appendix C - Risk Analysis

The potential dangers of the proposed system are listed in this appendix, along with corresponding mitigation techniques.

Table 6 - Risk Analysis

Risk	Description	Mitigation Strategy
Data availability	Some employers may not provide enough public information	Use multiple credibility indicators and a confidence indicator
Model bias	Dataset imbalance may affect model performance	Apply data preprocessing and model validation
Incorrect predictions	ML models may occasionally misclassify employers	Provide explanations and confidence indicators
System integration issues	Mobile app and backend communication errors	Implement API testing and error handling